Probability

A Practical, Systems-Based Approach

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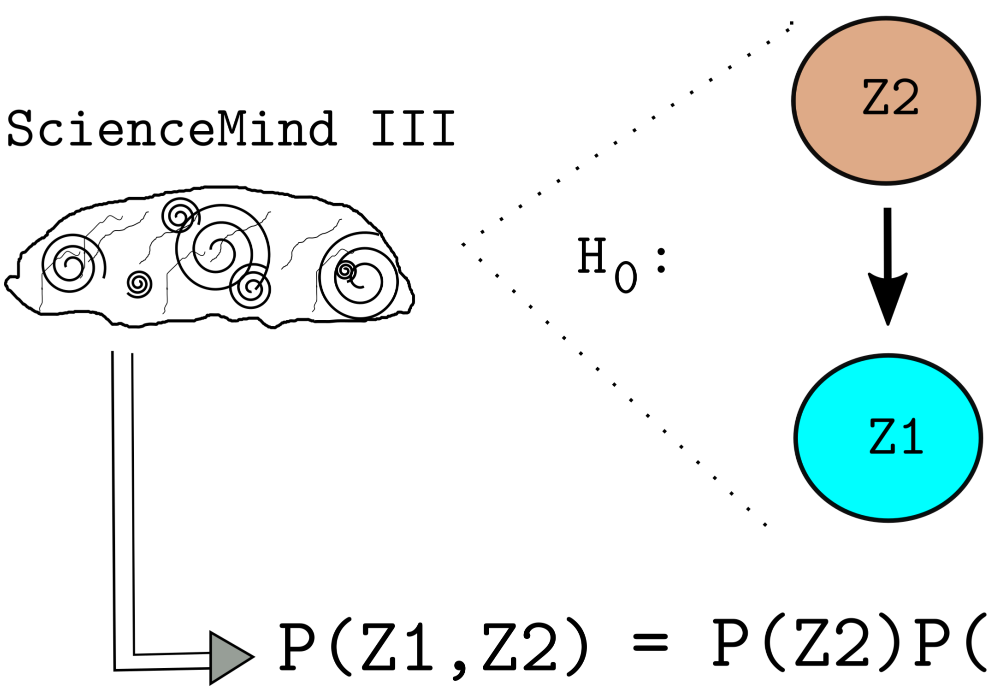
Z1={observed system behavior}
Z2={possible causally-linked data}



$$P(Z1,Z2) = P(Z2)P(Z1|Z2)$$

Preface

This book is dedeicated to hours of NOT understanding probability theory and applied statistics. For my part, I want a deeper understanding of the mechanisms of assumption and the mechanisms of application that arise in the study of both probability and statistics. I am awed by those excellent mathematical statisticians who can rapidly manipulate the requisite symbologies to acheive end results. But I am often left thinking that in the swirl of manipulations there are hidden assumptions of which I am unaware; there are prerequisites and extensions that I am missing. I am forever grateful to Andrew ??? and Cosmo ??? (two of those excellent mathematical statisticians) who honestly report the limitations of statistical modeling and thus have encouraged me to dig deeper. I also owe a huge intellectual debt to E.T. Jaynes, Sir Harold Jefferies, and David J.C. MacKay for their use of probability theory as the logic of science, and to W. Ross Ashby for his tremendous insights on mechanism in observation via cybernetics.

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1.5 Introduction to Probability Models, Sheldon Ross

Ross Examples

Suppose that n independent trials, each of which results in any of m possible outcomes with respective probabilities p1 , . . . , pm , $\sum pi = 1$, are continually performed. Let X denote the number of trials needed until each outcome has occurred at least once. Define the equation for X=n and explain the reasoning.

Let outcomes be $1, \ldots, m$ with probabilities $p_1, \ldots, p_m$ (independent across trials). For $n < m$, $\mathbb{P}(X = n) = 0$. For $n \geq m$

$$\boxed{\mathbb{P}(X = n) = \sum_{\varnothing \neq S \subseteq \{1,\ldots,m\}} (-1)^{|S|+1} \Big(\sum_{j \in S} p_j\Big)\Big(1 - \sum_{j \in S} p_j\Big)^{n-1}}$$

Why this is true (reasoning)

$X = n$ means: after $n - 1$ trials at least one outcome is missing, and on trial n the **last** missing type appears for the **first** time. Equivalently, pick any nonempty set S of outcomes and consider the event that **all** outcomes in S have been missing up to time $n - 1$ and one of them appears at time n. The probability that no outcome from S appears in a given trial is $1 - \sum_{j \in S} p_j$. So the chance they are all missing for the first $n - 1$ trials and then one of them appears on trial n is

$$\Big(1 - \sum_{j \in S} p_j\Big)^{n-1} \Big(\sum_{j \in S} p_j\Big).$$

* But these events for different S overlap (inclusion–exclusion fixes the overcount), giving the alternating-sum formula above.

Coupon Collector Cheat Sheet (General Probabilities)

Problem: m outcomes with probabilities $p_1, \ldots, p_m > 0$, $\sum p_i = 1$.
X = number of trials until **all outcomes appear at least once**.
Define $q_S = \sum_{j \in S} p_j$ for subset $S \subseteq \{1, \ldots, m\}$.

Core Formulas

Quantity	Formula
PMF ($n \geq m$)	$\mathbb{P}(X = n) = \sum_{\varnothing \neq S} (-1)^{\lvert S \rvert + 1} q_S (1 - q_S)^{n-1}$
CDF	$\mathbb{P}(X \leq n) = \sum_{S} (-1)^{\lvert S \rvert} (1 - q_S)^n$
Expectation	$\mathbb{E}[X] = \sum_{\varnothing \neq S} (-1)^{\lvert S \rvert + 1} \frac{1}{q_S}$
Minimum trials	$X_{\min} = m$
Sanity Check	$\mathbb{P}(X = m) = m! \prod p_i$; $\sum_{n \geq m} \mathbb{P}(X = n) = 1$

Quick Insights

- Tail behavior dominated by largest $1 - q_{\{i\}} = 1 - \min p_i$; decays geometrically.
- Uniform probs: $p_i = 1/m \Rightarrow \mathbb{E}[X] = mH_m \approx m(\ln m + \gamma)$.
- Complexity: exact computation $O(2^m)$; feasible for $m \lesssim 20$.
- Simulation: use Monte Carlo for large m.
- First m trials all distinct probability: $m! \prod p_i$.

Algorithm (Bitmask / Combinations)

1. Enumerate all non-empty subsets S (bitmask or itertools.combinations).
2. Compute $q_S = \sum_{j \in S} p_j$.
3. Plug into PMF or expectation formulas.
4. Verify probabilities sum to 1 within tolerance.

Python Helper

```python
from itertools import combinations

def coupon_collector_pmf(p, n):
    m = len(p)
    prob = 0.0
    idx = list(range(m))
    for r in range(1, m+1):
        for subset in combinations(idx, r):
            q = sum(p[i] for i in subset)
            prob += ((-1)**(r+1)) * q * (1-q)**(n-1)
    return prob

def coupon_collector_expectation(p):
    m = len(p)
    exp_val = 0.0
    idx = list(range(m))
```

```
    for r in range(1, m+1):
        for subset in combinations(idx, r):
            q = sum(p[i] for i in subset)
            exp_val += ((-1)**(r+1)) * (1/q)
    return exp_val

# Example usage:
p = [0.2, 0.3, 0.5]
for n in range(3, 8):
    print(f"P(X={n})␣=␣{coupon_collector_pmf(p,␣n):.5f}")
print("E[X]␣=", coupon_collector_expectation(p))
```

1.5.1 Example 2.5, p. 23

1.5.2 Excessive Elaborations

1.6 Ross Exercises Chapter 1

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